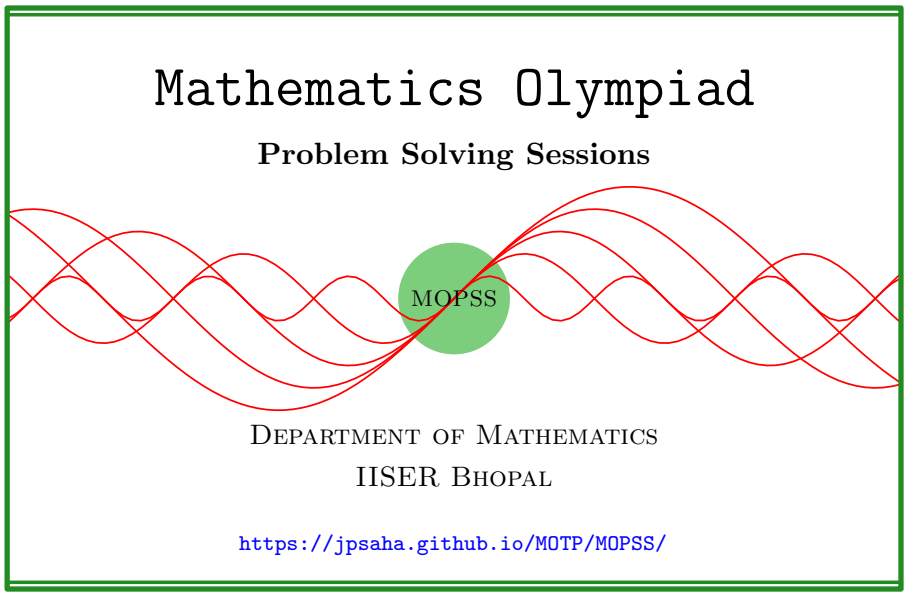
# Polynomials

MOPSS




## Suggested readings

- Evan Chen's advice On reading solutions, available at https://blog.evanchen.cc/2017/03/06/on-reading-solutions/.
- Evan Chen's Advice for writing proofs/Remarks on English, available at https://web.evanchen.cc/handouts/english/english.pdf.
- Notes on proofs by Evan Chen from OTIS Excerpts [**Che25**, Chapter 1].
- Tips for writing up solutions by Edward Barbeau, available at https://www.math.utoronto.ca/barbeau/writingup.pdf.
- Evan Chen discusses why math olympiads are a valuable experience for high schoolers in the post on Lessons from math olympiads, available at https://blog.evanchen.cc/2018/01/05/lessons-from-math-olympiads/.

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# §1 Polynomials

For further problems, we refer to [Goy21].

## §1.1 Warm up

**Example 1.** Factorize the polynomial $x^8 + x^4 + 1$ into factors of at most the second degree.

**Summary** — Expressing an expression as a difference of two squares yields a factorization.

**Solution 1.** Note that

$$\begin{aligned}
&x^8 + x^4 + 1 \\
&= x^8 + 2x^4 + 1 - x^4 \\
&= (x^4 + 1)^4 - (x^2)^2 \\
&= (x^4 - x^2 + 1)(x^4 + 1 + x^2) \\
&= (x^4 + 2x^2 + 1 - 3x^2)(x^4 + 2x^2 + 1 - x^2) \\
&= ((x^2 + 1)^2 - (\sqrt{3}x)^2)((x^2 + 1)^2 - x^2) \\
&= (x^2 - \sqrt{3}x + 1)(x^2 + \sqrt{3}x + 1)(x^2 - x + 1)(x^2 + x + 1).
\end{aligned}$$

■

**Example 2.** Show that

$$2a^2b^2 + 2b^2c^2 + 2c^2a^2 - a^4 - b^4 - c^4 = (a+b+c)(a+b-c)(b+c-a)(c+a-b).$$

**Summary** — Complete the squares.

**Walkthrough** —

**(a)** Try to see what would happen if we were allowed to change the signs!

**(b)** Change the signs and consider

$$a^4 + b^4 + c^4 - 2a^2b^2 - 2b^2c^2 - 2c^2a^2.$$

This is "almost" $(a^2 - b^2 - c^2)^2$!! To be precise

$$(a^2 - b^2 - c^2)^2 = a^4 + b^4 + c^4 - 2a^2b^2 + 2b^2c^2 - 2c^2a^2.$$

**(c)** Let us continue with the above, and write $2a^2b^2+2b^2c^2+2c^2a^2-a^4-b^4-c^4$ in terms $(a^2 - b^2 - c^2)^2$ as follows.

$$\begin{aligned}
&2a^2b^2 + 2b^2c^2 + 2c^2a^2 - a^4 - b^4 - c^4 \\
&= 4b^2c^2 - (a^4 + b^4 + c^4 - 2a^2b^2 + 2b^2c^2 - 2c^2a^2).
\end{aligned}$$

**(d)** Does the above help?

**Solution 2.** Note that

$$2a^2b^2 + 2b^2c^2 + 2c^2a^2 - a^4 - b^4 - c^4$$

$$
\begin{aligned}
&= 4b^2c^2 - (a^4 + b^4 + c^4 - 2a^2b^2 + 2b^2c^2 - 2c^2a^2) \\
&= (2bc)^2 - (a^2 - b^2 - c^2)^2 \\
&= (2bc - (a^2 - b^2 - c^2))(2bc + a^2 - b^2 - c^2) \\
&= (2bc + b^2 + c^2 - a^2)(a^2 - (b^2 + c^2 - 2bc)) \\
&= ((b + c)^2 - a^2)(a^2 - (b - c)^2) \\
&= (a + b + c)(b + c - a)(a + b - c)(a - b + c) \\
&= (a + b + c)(a + b - c)(b + c - a)(c + a - b).
\end{aligned}
$$

■

**Solution 3.** Write[1]

$$
\begin{aligned}
a &= y + z, \\
b &= z + x, \\
c &= x + y.
\end{aligned}
$$

Note that

$$
\begin{aligned}
&2a^2b^2 + 2b^2c^2 + 2c^2a^2 - a^4 - b^4 - c^4 \\
&= 2(y + z)^2(z + x)^2 + 2(z + x)^2(x + y)^2 + 2(x + y)^2(y + z)^2 \\
&- (y + z)^4 - (z + x)^4 - (x + y)^4 \\
&= 2(z^2 + 2yz + y^2)(z^2 + 2zx + x^2) \\
&+ 2(x^2 + 2zx + z^2)(x^2 + 2xy + y^2) \\
&+ 2(y^2 + 2xy + x^2)(y^2 + 2yz + z^2) \\
&- (y + z)^4 - (z + x)^4 - (x + y)^4 \\
&= 2\sum_{\text{cyc}} \left(x^4 + x^2(y^2 + z^2 + 2x(y + z)) + yz(2x + y)(2x + z)\right) - \sum_{\text{cyc}} (x + y)^4 \\
&= 2\sum_{\text{cyc}} \left(x^4 + x^2y^2 + z^2x^2 + 2x^3(y + z) + 4x^2yz + 2xyz(y + z) + y^2z^2\right) \\
&- \sum_{\text{cyc}} (x + y)^4 \\
&= 2(x^4 + y^4 + z^4) + 6(x^2y^2 + y^2z^2 + z^2x^2) + 16xyz(x + y + z) + 4\sum_{\text{cyc}} x^3(y + z)
\end{aligned}
$$

[1] It is good ask the following simple and innocent question: how can one **write** $a, b, c$ as stated above? Does it mean that given any three real numbers $a, b, c$, one can find real numbers $x, y, z$ such that $a = y + z, b = z + x, c = x + y$? A crucial point to note is that one very often deals with indeterminates (aka variables) instead of real numbers. In the above, $a, b, c$ could be indeterminates instead of being real numbers! What do we do in that case? Is it so that there are **indeterminates** $x, y, z$ such that the six indeterminates $a, b, c, x, y, z$ satisfy $a = y + z, b = z + x, c = x + y$? As of now, **let's not worry about any of these!**. Just keep in mind the message that at times, **we need to be quite careful about what we do!**

$$
\begin{aligned}
&-\sum_{\text{cyc}}(x+y)^4 \\
&= 2(x^4+y^4+z^4)+6(x^2y^2+y^2z^2+z^2x^2)+16xyz(x+y+z)+4\sum_{\text{cyc}}x^3(y+z) \\
&-\sum_{\text{cyc}}(x^4+4x^3y+6x^2y^2+4xy^3+y^4) \\
&= 16xyz(x+y+z) \\
&= (a+b+c)(a+b-c)(b+c-a)(c+a-b).
\end{aligned}
$$

■

**Remark.** Please go through the footnote next to the word **Write** from the above solution. This footnote would explain that just writing **Write** $a=y+z, b=z+x, c=x+y$ requires more care! To address this issue, replace **Write** $a=y+z, b=z+x, c=x+y$ in the above solution by the following.

Consider the real numbers $x, y, z$ defined by

$$
\begin{aligned}
x &= \frac{1}{2}(b+c-a), \\
y &= \frac{1}{2}(c+a-b), \\
z &= \frac{1}{2}(a+b-c).
\end{aligned}
$$

Note that

$$a=y+z, b=z+x, c=x+y$$

holds.

**Example 3.** Find numbers $a, b, c, d$ for which the equation

$$\frac{2x-7}{4x^2+16x+15} = \frac{a}{x+c}+\frac{b}{x+d}$$

would be an identity.

**Walkthrough** — Factorize the denominator into linear factors. Then expressing the numerator as a linear combination of those factors would provide such an identity.

**Solution 4.** Note that

$$\frac{2x-7}{4x^2+16x+15} = \frac{2x-7}{(2x+3)(2x+5)}.$$

Hence, if $2x-7$ can be expressed as

$$p(2x+3)+q(2x+5),$$

then $\frac{2x-7}{4x^2+16x+15}$ can be expressed as a sum of two fractions, each having a constant in the numerator and a linear polynomial in the denominator.

One way to find if there are any such $p, q$, is to assume first that there are such real numbers $p$ and $q$ such that

$$2x - 7 = p(2x + 3) + q(2x + 5)$$

holds[2]. Substituting $x = -\frac{5}{2}$, we obtain $-2p = -12$, which gives $p = 6$. Next, substituting $x = -\frac{3}{2}$, we obtain $2q = -10$, which implies $q = -5$.

Note that

$$6(2x + 3) + (-5)(2x + 5) = 12x + 18 - 10x - 25 = 2x - 7$$

holds[3]. Using it, we obtain

$$\begin{aligned}\frac{2x-7}{4x^2+16x+15} &= \frac{2x-7}{(2x+3)(2x+5)}\\ &= \frac{6(2x+3)+(-5)(2x+5)}{(2x+3)(2x+5)}\\ &= \frac{6}{2x+5} - \frac{5}{2x+3}\\ &= \frac{3}{x+\frac{5}{2}} - \frac{\frac{5}{2}}{x+\frac{3}{2}}.\end{aligned}$$

Hence, we may take

$$a = 3, b = -\frac{5}{2}, c = \frac{5}{2}, d = \frac{3}{2}.$$

■

**Exercise 4.** Are there other choices for $a, b, c, d$ for which the identify would hold?

**Example 5.** Let $n$ be a positive integer. Show that

$$x^n - 1 = (x - 1)(x^{n-1} + x^{n-2} + \cdots + x + 1).$$

[2] and try to see what conditions get imposed on $p, q$. It may happen that the conditions that get imposed, may suggest that there are no such $p, q$. However, it may also happen that we would be able to find out which $p, q$ would work!

[3] It should be noted that $p, q$ were assumed to exist such that $p(2x+3)+q(2x+5) = 2x-7$ holds. Under this hypothesis, we obtained $p = 6, q = -5$. At this point, we cannot immediately conclude that $6(2x + 3) + (-5)(2x + 5) = 2x - 7$ holds (unless we verify it), because if we do so, then we would do it under the same hypothesis.

- Even then, what would go wrong with that?
- Can a hypothesis (possibly combined with some of its consequences) be a justification for itself to hold? Think about this point.



The content posted here and at this blog by Evan Chen are quite useful.

**Solution 5.** Note that

$$\begin{aligned}
&(x-1)(x^{n-1}+x^{n-2}+\cdots+x+1)\\
&=x(x^{n-1}+x^{n-2}+\cdots+x+1)-(x^{n-1}+x^{n-2}+\cdots+x+1)\\
&=x^n+x^{n-1}+\cdots+x^2+x-(x^{n-1}+x^{n-2}+\cdots+x+1)\\
&=x^n-1.
\end{aligned}$$

■

The exercise below relies on Example 5.

**Exercise 6** (Moscow Mathematical Olympiad 2015 Grade 9 P6)**.** Do there exist two polynomials with integer coefficients such that each of them has a coefficient with absolute value exceeding 2015, but no coefficient of their product has absolute value exceeding 1?

**Summary** — Try to come up with enough polynomials $g_1(x), g_2(x), g_3(x), \dots$ and $h_1(x), h_2(x), h_3(x), \dots$ such that each of the products $g_1g_2g_3\dots$ and $h_1h_2h_3\dots$ have at least one coefficient which is **large in absolute value**, and all the coefficients of the product $(g_1g_2g_3\dots)(h_1h_2h_3\dots)$ are at most 1 in absolute value.

**Walkthrough** —

**(a)** Try to come up with a polynomial $P(x)$ whose coefficients are at most 1 in absolute value, and it can be written as a product of enough factors (say $f_1(x), f_2(x), \dots$) such that each of such factor $f_i(x)$ admits a decomposition into the product of two polynomials $g_i(x)$ and $h_i(x)$.

**(b)** Can you make sure that the product of the $g_i$'s, and the product of the $h_i$'s have to have at least one large coefficient?

**(c)** For instance, would taking $g_1(x)=g_2(x)=g_3(x)=\cdots=1-x$ work for some suitable choice of $h_1(x), h_2(x), \dots$?

**(d)** Does taking

$$\begin{aligned}
h_1(x)&=1+x,\\
h_2(x)&=1+x+x^2,\\
h_3(x)&=1+x+x^2+x^3,
\end{aligned}$$

etc. work?

**(e)** Note that the product of enough $g_i$'s would have a large coefficient (namely, the coefficient of the second largest power of $x$). On the other hand, the product of enough $h_i$'s would have a large coefficient (namely, the coefficient of the power of $x$).

**(f)** What can be said about the absolute value of the coefficients of the product of these two products?

The above seems to work except that having a control on the coefficients of the product $(g_1g_2g_3\dots)(h_1h_2h_3\dots)$ seems hard[4].

**Solution 6.** Consider the polynomial

$$P(x) = (1-x)(1-x^2)(1-x^4)(1-x^8)\cdots(1-x^{2^{2016}}).$$

Since

$$1+2+2^2+2^3+\cdots+2^{n-1} < 2^n,$$

it follows that the coefficients of $P(x)$ are at most 1 in absolute value. Note that

$$P(x) = Q(x)R(x)$$

holds where

$$\begin{aligned} Q(x) &= (1-x)^{2017}, \\ R(x) &= (1+x)(1+x+x^2+x^3)\cdots(1+x+x^2+\cdots+x^{2^{2016}-1}). \end{aligned}$$

The coefficient of $x^{2016}$ in $Q(x)$ is equal to 2017, and the coefficient of $x$ in $R(x)$ is equal to 2016. This completes the proof. ■

**Exercise 7** (All-Russian Mathematical Olympiad 2007 Grade 10 Day 1 P2, , by A. Khrabrov)**.** Consider a polynomial $P(x) = a_0x^n + a_1x^{n-1} + \cdots + a_{n-1}x + a_n$. Let $m$ denote the minimum of the integers

$$a_0, a_0+a_1, \dots, a_0+a_1+\cdots+a_n.$$

Prove that $P(x) \geq mx^n$ for all $x \geq 1$.

**Walkthrough** —

**(a)**

**Solution 7.** Note that

$$\begin{aligned} P(x) &= a_0x^n + a_1x^{n-1} + \cdots + a_{n-1}x + a_n \\ &= a_0(x^n - x^{n-1}) + (a_0+a_1)(x^{n-1}-x^{n-2}) \\ &\quad + \cdots + (a_0+a_1+\cdots+a_{n-1})(x-1) + (a_0+a_1+\cdots+a_n) \end{aligned}$$

holds, which implies that

$$P(x) - mx^n$$

[4] Is it because it fails?

$$
\begin{aligned}
&= a_0(x^n - x^{n-1}) + (a_0 + a_1)(x^{n-1} - x^{n-2}) \\
&\quad + \cdots + (a_0 + a_1 + \cdots + a_{n-1})(x - 1) + (a_0 + a_1 + \cdots + a_n) \\
&\quad - m(x^n - x^{n-1}) - m(x^{n-1} - x^{n-2}) - \cdots - m(x - 1) - m \\
&= (a_0 - m)(x^n - x^{n-1}) + (a_0 + a_1 - m)(x^{n-1} - x^{n-2}) \\
&\quad + \cdots + (a_0 + a_1 + \cdots + a_{n-1} - m)(x - 1) + (a_0 + a_1 + \cdots + a_n - m).
\end{aligned}
$$

It follows that for any $x \geq 1$,

$$P(x) \geq mx^n.$$

■

## §1.2 Division algorithm

**Example 8.** Prove that the polynomial $x^{44} + x^{33} + x^{22} + x^{11} + 1$ is divisible by the polynomial $x^4 + x^3 + x^2 + x + 1$.

**Solution 8.** Note that

$$
\begin{aligned}
&x^{44} + x^{33} + x^{22} + x^{11} + 1 \\
&= x^{40} \cdot x^4 + x^{30} \cdot x^3 + x^{20} \cdot x^2 + x^{10} \cdot x + 1 \\
&= (x^{40} - 1)x^4 + (x^{30} - 1)x^3 + (x^{20} - 1)x^2 + (x^{10} - 1)x \\
&\quad + x^4 + x^3 + x^2 + x + 1.
\end{aligned}
$$

Hence, to prove that the polynomial $x^{44} + x^{33} + x^{22} + x^{11} + 1$ is divisible by $x^4 + x^3 + x^2 + x + 1$, it suffices to show that $x^4 + x^3 + x^2 + x + 1$ divides the polynomials

$$x^{40} - 1, x^{30} - 1, x^{20} - 1, x^{10} - 1.$$

Since

$$
\begin{aligned}
x^5 - 1 &= (x - 1)(x^4 + x^3 + x^2 + x + 1), \\
x^{10} - 1 &= (x^5 - 1)(x^5 + 1),
\end{aligned}
$$

it follows that $x^4 + x^3 + x^2 + x + 1$ divides $x^{10} - 1$. Moreover, the polynomial $x^{10-1}$ divides all of

$$x^{40} - 1, x^{30} - 1, x^{20} - 1, x^{10} - 1.$$

Hence, $x^4 + x^3 + x^2 + x + 1$ divides the polynomials

$$x^{40} - 1, x^{30} - 1, x^{20} - 1, x^{10} - 1.$$

■

**Exercise 9.** Show that the polynomial $x^{580}+x^{390}+x^{326}+x^{262}+x^{198}+x^{134}+1$ is divisible by $x^6+x^5+x^4+x^3+x^2+x+1$.

**Example 10.** Determine the remainder obtained upon dividing $x^{100}$ by $x^2-3x+2$.

**Solution 9.** Let $q(x)$ (resp. $r(x)$) denote the quotient (resp. the remainder) obtained upon dividing $x^{100}$ by $x^2-3x+2$. Note that $r(x)$ is a linear polynomial, i.e. $r(x)=ax+b$ for some real numbers $a,b$. Then we have

$$x^{100}=q(x)(x^2-3x+2)+r(x).$$

Substituting $x=1$, it yields

$$1=r(1)=a+b.$$

Similarly, substituting $x=2$, it gives

$$2^{100}=r(2)=2a+b.$$

This shows that

$$a=2^{100}-1,\quad b=1-a=2-2^{100}.$$

Hence, the remainder obtained upon dividing $x^{100}$ by $x^2-3x+2$ is equal to

$$(2^{100}-1)x+2-2^{100}.$$

■

**Example 11.** Factorize the polynomials

$$x(y-z)^3+y(z-x)^3+z(x-y)^3,\quad xy(x^2-y^2)+yz(y^2-z^2)+zx(z^2-x^2)$$

into products of linear polynomials.

## §1.3 Bézout's theorem

**Example 12.** Let $f(x),g(x)$ be polynomials with integer coefficients and with no common factors. Show that $\gcd(f(n),g(n))$ attains only finitely many values as $n$ varies over the integers.

## §1.4 Even and odd polynomials

**Exercise 13** (Moscow Mathematical Olympiad First Round 1946 Grades 7-8 P5)**.** Prove that after completing the multiplication and collecting the terms

$$(1-x+x^2-x^3+\cdots-x^{99}+x^{100})(1+x+x^2+\cdots+x^{99}+x^{100})$$

has no monomials of odd degree.



The content posted here and at this blog by Evan Chen are quite useful.

**Summary** — What happens if $x$ is replaced by $-x$?

**Walkthrough** —

**(a)**

**Solution 10.** Let $P(x)$ denote the polynomial

$$(1 - x + x^2 - x^3 + \cdots - x^{99} + x^{100})(1 + x + x^2 + \cdots + x^{99} + x^{100}).$$

Note that $P(x) = P(-x)$. By the Claim below, it follows that $P(x)$ has no monomials of odd degree.

**Claim** — Let $Q(x)$ be a polynomial satisfying $Q(x) = Q(-x)$. Then $Q(x)$ has no monomials of odd degree.

*Proof of the Claim.* Note that

$$Q(x) = \frac{Q(x) + Q(-x)}{2} + \frac{Q(x) - Q(-x)}{2}$$

holds. Using $Q(x) = Q(-x)$, it follows that $Q(x) = \frac{Q(x)+Q(-x)}{2}$. Consequently, $Q(x)$ has no monomials of odd degree. □

■

**Remark.** The above decomposition of $Q(x)$ is a special case of general phenomena[a].

[a] Can you think of a few? Which **general phenomena** is referred to?!

**Remark.** The above solution is more elegant, and less cumbersome. Moreover, it also highlights the underlying reason, whereas the solution below obscures the conceptual viewpoint.

**Solution 11.** One can multiply the polynomials to note that

$$\begin{aligned}
&1 - x + x^2 - x^3 + \cdots - x^{99} + x^{100} \\
&= 1 - x + x^2(1 - x) + x^4(1 - x) + x^6(1 - x) + \cdots + x^{98}(1 - x) + x^{100} \\
&= (1 - x)(1 + x^2 + x^4 + x^6 + \cdots + x^{98}) + x^{100}.
\end{aligned}$$

Using this, we obtain

$$\begin{aligned}
&(1 - x + x^2 - x^3 + \cdots - x^{99} + x^{100})(1 + x + x^2 + \cdots + x^{99} + x^{100}) \\
&= \big((1 - x)(1 + x^2 + x^4 + x^6 + \cdots + x^{98}) + x^{100}\big)(1 + x + x^2 + \cdots + x^{99} + x^{100}) \\
&= (1 - x)(1 + x^2 + x^4 + x^6 + \cdots + x^{98})(1 + x + x^2 + \cdots + x^{99} + x^{100})
\end{aligned}$$

$$
\begin{aligned}
&\quad + x^{100}(1 + x + x^2 + \cdots + x^{99} + x^{100}) \\
&= (1 + x^2 + x^4 + x^6 + \cdots + x^{98})(1 - x^{101}) \\
&\quad + x^{100}(1 + x + x^2 + \cdots + x^{99} + x^{100}) \\
&= (1 + x^2 + x^4 + x^6 + \cdots + x^{98})(1 - x^{101}) \\
&\quad + x^{100}(x + x^3 + x^5 + \cdots + x^{99}) \\
&\quad + x^{100}(1 + x^2 + x^4 + \cdots + x^{98} + x^{100}) \\
&= (1 + x^2 + x^4 + x^6 + \cdots + x^{98})(1 - x^{101}) \\
&\quad + x^{101}(1 + x^2 + x^4 + x^6 + \cdots + x^{98}) \\
&\quad + x^{100}(1 + x^2 + x^4 + \cdots + x^{98} + x^{100}) \\
&= 1 + x^2 + x^4 + x^6 + \cdots + x^{98} + x^{100}(1 + x^2 + x^4 + \cdots + x^{98} + x^{100}),
\end{aligned}
$$

which has no monomial of odd degree. ■

The following exercise is quite similar to the Claim proved in the solution to **??**.

**Exercise 14.** Let $Q(x)$ be a polynomial satisfying $Q(x) = -Q(-x)$. Then $Q(x)$ has no monomials of even degree.

**Example 15.** Let $n$ be an even positive integer, and let $p(x)$ be a polynomial of degree $n$ such that $p(k) = p(-k)$ for $k = 1, 2, \ldots, n$. Prove that there is a polynomial $q(x)$ such that $p(x) = q(x^2)$.

**Walkthrough** —

**(a)** Note that the polynomial $p(x) - p(-x)$ has degree $< n$ because $n$ is even. Observe that it has at least $n$ roots.

**Remark.** What would happen if $n$ is not assumed to be even?

**Solution 12.** Note that the polynomial $p(x) - p(-x)$ has degree at most $n$, and it vanishes at the integers $-n, \ldots, -2, 1, 0, 1, 2, \ldots, n$. Thus, it has at least $2n + 1$ roots. It follows that the polynomial is identically zero, that is, the polynomials $p(x), p(-x)$ are equal. This implies that $p(x)$ is equal to $q(x^2)$ for some polynomial $q(x)$. ■

**Exercise 16** (Tournament of Towns Spring 2014, Senior A Level P7, by D. A. Zvonkin)**.** Consider a polynomial $P(x)$ such that

$$P(0) = 1, \quad (P(x))^2 = 1 + x + x^{100}Q(x),$$

where $Q(x)$ is also a polynomial. Prove that in the polynomial $(P(x) + 1)^{100}$, the coefficient of $x^{99}$ is zero.

**Walkthrough** —

**(a)** Since $P(x)^2$ is congruent to $1+x$ modulo $x^{100}$, show that $(P(x)+1)^{100}+(1-P(x))^{100}$ is congruent to a polynomial of degree 50 in $1+x$ modulo $x^{100}$.

**(b)** Prove that $(P(x)+1)^{100}$ is congruent to a polynomial of degree 50 in $1+x$ modulo $x^{100}$.

**Solution 13.** Note that

$$(P(x)+1)^{100}+(1-P(x))^{100}$$

is a polynomial in $P(x)^2$ of degree 50. Given three polynomials $f(x), g(x), h(x)$ having complex coefficients, with $h(x) \neq 0$, we say that $f(x)$ **is congruent to** $g(x)$ **modulo** $h(x)$ if $h(x)$ divides $f(x)-g(x)$, that is, $f(x)-g(x)$ is the product of $h(x)$ and a polynomial in $x$ with complex coefficients. Since $P(x)^2$ is congruent to $1+x$ modulo $x^{100}$, it follows that $(P(x)+1)^{100}+(1-P(x))^{100}$ is congruent to a polynomial of degree 50 in $1+x$ modulo $x^{100}$. Using that $P(x) \equiv 1 \bmod x$, we obtain that $(P(x)+1)^{100}$ is congruent to a polynomial of degree 50 in $1+x$ modulo $x^{100}$. This shows that the coefficient of $x^{99}$ in $(P(x)+1)^{100}$ is zero. ■

## §1.5 Factorization and roots

**Example 17.** Let $a, b, c$ be three distinct real numbers. Show that

$$\frac{(a-x)(b-x)}{(a-c)(b-c)}+\frac{(b-x)(c-x)}{(b-a)(c-a)}+\frac{(c-x)(a-x)}{(c-b)(a-b)}=1.$$

**Walkthrough** — Can a polynomial having degree at most two admit more than two distinct roots?

**Exercise 18** (All-Russian Mathematical Olympiad 2011 Grade 9 Day 1 P1, , by A. Khrabrov)**.** Let $P(x)$ be a monic quadratic polynomial such that $P(x)$ and $P(P(P(x)))$ admit a common root. Prove that $P(0) \cdot P(1) = 0$.

**Walkthrough** —

**(a)** Show that $P(P(0))$ is equal to 0.

**(b)** Show that $P(P(0)) = P(0) \cdot P(1)$.

**Solution 14.** Let $\alpha$ denote a common root of the polynomials $P(x)$, $P(P(P(x)))$. Using

$$P(P(P(\alpha)))=P(\alpha)=0,$$

it follows that

$$P(P(0)) = 0.$$

Since $P(x)$ is a monic quadratic polynomial, we obtain that

$$P(P(0)) = P(0) \cdot P(1).$$

This shows that $P(0) \cdot P(1) = 0$. ■

**Exercise 19** (USAMO 1975 P3, ). A polynomial $P(x)$ of degree $n$ satisfies

$$P(k) = \frac{k}{k+1} \quad \text{for } k = 0, 1, 2, \ldots, n.$$

Find $P(n+1)$.

**Walkthrough** —

**(a)** Consider the polynomial $(x+1)P(x) - x$.

**Solution 15.** Note that $xP(x+1) - x$ is a polynomial of degree $n+1$, and it vanishes at the $n+1$ integers $0, 1, 2, \ldots, n$. It follows that

$$(x+1)P(x) - x = cx(x-1)(x-2)\ldots(x-n)$$

for some nonzero real number $c$. Substituting $x = -1$ yields

$$-1 = (-1)^{n+1} c(n+1)!,$$

which gives $c = \frac{(-1)^n}{(n+1)!}$. This implies that

$$(n+2)P(n+1) = n + 1 + (-1)^n,$$

and consequently,

$$P(n+1) = \frac{n+1+(-1)^n}{n+2}.$$

■

**Exercise 20** (International Mathematics Competition 2008 Day 2 P4). Let $f(x), g(x)$ be nonconstant polynomials with integer coefficients such that $g(x)$ divides $f(x)$. Prove that if the polynomial $f(x) - 2008$ has at least 81 distinct integer roots, then the degree of $g(x)$ is greater than 5.

**Walkthrough** —

**(a)**

**Solution 16.** Without loss of generality, we may assume that $g(x)$ has content 1, that is, the greatest common divisor of its coefficients is 1. Since $g(x)$ divides $f(x)$, there exists a polynomial $h(x)$ with rational coefficients such that $f(x) = g(x)h(x)$. Since $g(x)$ is a primitive polynomial with integer coefficients, by Gauss's lemma, the polynomial $h(x)$ also has integer coefficients. Denote the set of integer roots of $f(x) - 2008$ by $S$. Note that for each $a \in S$, we have $f(a) = 2008$, and thus $g(a)$ divides 2008. Since $2008 = 2^3 \cdot 251$, the number of positive integer divisors of 2008 is 8. Therefore, the polynomial $g(x)$ can take at most 16 distinct integer values on the set $S$. By hypothesis, the set $S$ has at least 81 elements. By the pigeonhole principle, there exists a divisor $d$ of 2008 such that the equation $g(x) = d$ has at least $\lceil 81/16 \rceil = 6$ distinct integer solutions. This shows that the polynomial $g(x) - d$ has at least 6 distinct integer roots, and hence its degree is at least 6. This implies that the degree of $g(x)$ is greater than 5, as required. ■

**Exercise 21** (Asian Pacific Mathematics Olympiad 2009 P2, ). Let $a_1, a_2, a_3, a_4, a_5$ be real numbers satisfying the following equations:

$$\frac{a_1}{k^2+1} + \frac{a_2}{k^2+2} + \frac{a_3}{k^2+3} + \frac{a_4}{k^2+4} + \frac{a_5}{k^2+5} = \frac{1}{k^2}$$

for $k = 1, 2, 3, 4, 5$. Find the value of

$$\frac{a_1}{37} + \frac{a_2}{38} + \frac{a_3}{39} + \frac{a_4}{40} + \frac{a_5}{41}.$$

(Express the value in a single fraction.)

**Solution 17.** Consider the polynomial

$$\begin{aligned} P(x) = {} & a_1 x(x+2)(x+3)(x+4)(x+5) \\ & + a_2 x(x+1)(x+3)(x+4)(x+5) \\ & + a_3 x(x+1)(x+2)(x+4)(x+5) \\ & + a_4 x(x+1)(x+2)(x+3)(x+5) \\ & + a_5 x(x+1)(x+2)(x+3)(x+4) \\ & - (x+1)(x+2)(x+3)(x+4)(x+5). \end{aligned}$$

Note that $P(x)$ is a polynomial of degree at most 5, and it vanishes at $1^2, 2^2, 3^2, 4^2, 5^2$. Thus, we can write

$$P(x) = c(x-1)(x-4)(x-9)(x-16)(x-25)$$

for some real number $c$. Note that

$$P(0) = -120.$$

On the other hand,

$$\begin{aligned} P(0) &= c(-1)(-4)(-9)(-16)(-25) \\ &= -14400c. \end{aligned}$$

Equating the two expressions for $P(0)$, we get

$$c = \frac{1}{120}.$$

Note that

$$\begin{aligned} S &= \frac{a_1}{37} + \frac{a_2}{38} + \frac{a_3}{39} + \frac{a_4}{40} + \frac{a_5}{41} \\ &= \frac{P(36)}{36 \cdot 37 \cdot 38 \cdot 39 \cdot 40 \cdot 41} + \frac{1}{36} \\ &= \frac{1}{120} \times \frac{(36-1)(36-4)(36-9)(36-16)(36-25)}{36 \cdot 37 \cdot 38 \cdot 39 \cdot 40 \cdot 41} + \frac{1}{36} \\ &= \frac{1}{120} \times \frac{35 \cdot 32 \cdot 27 \cdot 20 \cdot 11}{36 \cdot 37 \cdot 38 \cdot 39 \cdot 40 \cdot 41} + \frac{1}{36} \\ &= \frac{55440}{36 \cdot 37 \cdot 38 \cdot 39 \cdot 40 \cdot 41} + \frac{37 \cdot 38 \cdot 39 \cdot 40 \cdot 41}{36 \cdot 37 \cdot 38 \cdot 39 \cdot 40 \cdot 41} \\ &= \frac{55440 + 89927760}{36 \cdot 37 \cdot 38 \cdot 39 \cdot 40 \cdot 41} \\ &= \frac{89983200}{36 \cdot 37 \cdot 38 \cdot 39 \cdot 40 \cdot 41} \\ &= \frac{187465}{6744582}. \end{aligned}$$

■

**Exercise 22** (Belarus National Olympiad 2025 Grade 10 Day 1 P2, , by A. Voidelevich)**.** Let $n$ be a positive integer and $P(x)$ be a polynomial with integer coefficients such that

$$P(1) = 1, P(2) = 2, \ldots, P(n) = n.$$

Prove that $P(0)$ is divisible by $2 \cdot 3 \cdot \ldots \cdot n$.

**Walkthrough** —

**(a)** Are the polynomials $x$ and $(1-x)(2-x)(3-x)\cdots(n-x)$ of some use?

**(b)** Does division algorithm help?

**Solution 18.** Since $(1-x)(2-x)(3-x)\cdots(n-x)$ is polynomial with integer coefficients and its leading coefficient is $(-1)^n$, it follows by induction that there exist polynomials $Q(x)$ and $R(x)$ with integer coefficients such that

$$P(x) - x = Q(x)(1-x)(2-x)(3-x)\cdots(n-x) + R(x),$$

where $R(x)$ is the zero polynomial, or the degree of $R(x)$ is less than $n$. Note that $R(k) = P(k) - k = 0$ for all $k = 1, 2, \ldots, n$. Thus, $R(x)$ is the zero polynomial. Therefore,

$$P(x) - x = Q(x)(1-x)(2-x)(3-x)\cdots(n-x).$$

Setting $x = 0$ implies that $P(0)$ is divisible by $2 \cdot 3 \cdot \ldots \cdot n$. ■

**Example 23.** Let $P(x)$ be a polynomial of degree $\leq n$ having rational coefficients. Suppose $P(k) = \frac{1}{k}$ holds for $1 \leq k \leq n+1$. Determine $P(0)$.

**Solution 19.** Note that $xP(x) - 1$ is a polynomial of degree at most $n+1$, and it vanishes at $1, 2, \ldots, n+1$. This implies that

$$xP(x) - 1 = c(x-1)(x-2)\ldots(x-n-1)$$

holds for some rational number $c$. Substituting $x = 0$, we obtain

$$c = \frac{(-1)^n}{(n+1)!}.$$

Differentiating and dividing by $xP(x) - 1$, we obtain

$$\frac{xP'(x) + P(x)}{xP(x) - 1} = \frac{1}{x-1} + \frac{1}{x-2} + \cdots + \frac{1}{x-n-1},$$

which yields

$$P(0) = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}.$$

■

**Example 24.** Let $g(x)$ and $h(x)$ be polynomials with real coefficients such that

$$g(x)(x^2 - 3x + 2) = h(x)(x^2 + 3x + 2)$$

and $f(x) = g(x)h(x) + (x^4 - 5x^2 + 4)$. Prove that $f(x)$ has at least four real roots.

**Solution 20.** Note that $g(x)$ and $h(x)$ satisfy

$$g(x)(x-1)(x-2) = h(x)(x+1)(x+2),$$

which shows that

$$g(-1), g(-2), h(1), h(2)$$

are equal to 0. Also note that

$$\begin{aligned} x^4 - 5x^2 + 4 &= (x^2-1)(x^2-4) \\ &= (x-1)(x+1)(x+2)(x-2). \end{aligned}$$

Hence, the polynomials $g(x)h(x)$ and $x^4 - 5x^2 + 4$ vanish at $1, -1, 2, -2$. Consequently, $f$ also vanishes at these four points. ■

**Example 25.** Let $P(x)$ be a polynomial with real coefficients such that $P(\sin\alpha) = P(\cos\alpha)$ for all $\alpha \in \mathbb{R}$. Show that $P(x) = Q(x^2 - x^4)$ for some polynomial $Q(x)$ with real coefficients.

**Walkthrough** —

**(a)** Show that $P(x) = P(-x)$ for any $-1 \le x \le 1$, and hence $P(x) = f(x^2)$.

**(b)** Deduce that $f(x) = f(1-x)$ for any $0 \le x \le 1$.

**(c)** Using induction or otherwise, prove that $f(x) = g(x - x^2)$ for some polynomial $g(x)$ with real coefficients.

**Exercise 26** (BMath 2006 P3)**.** [cf. Moscow MO 1953 Grade 10, India BMath 2006 P3] [**AE11**, §1.7] Let $n$ be a positive integer. Find the roots of the polynomial

$$P_n(X) = 1 + \frac{X}{1!} + \frac{X(X+1)}{2!} + \cdots + \frac{X(X+1)\cdots(X+n-1)}{n!}.$$

**Walkthrough** — Note that $P_1(X) = X + 1$ has $-1$ as its root, $P_2(X) = \frac{1}{2}(X+1)(X+2)$ has $-1, -2$ as its roots. Check that

$$P_3(X) = \frac{1}{3!}(X+1)(X+2)(X+3).$$

What happens for general $n$?

**Solution 21.** We claim that the roots of $P_n(X)$ are $-1, -2, \ldots, -n$ for any integer $n \ge 1$. Note that the claim holds for $n = 1$. Suppose the claim holds for some integer $n$. Comparing leading coefficients, it follows that

$$P_n(X) = \frac{1}{n!}(X+1)(X+2)\cdots(X+n).$$

Observe that

$$\begin{aligned} P_{n+1}(X) &= P_n(X) + \frac{X(X+1)\cdots(X+n)}{(n+1)!} \\ &= \frac{1}{n!}(X+1)(X+2)\cdots(X+n) + \frac{X(X+1)\cdots(X+n)}{(n+1)!} \\ &= \frac{1}{(n+1)!}(X+1)(X+2)\cdots(X+n)(n+1+X). \end{aligned}$$

Hence the roots of $P_{n+1}$ are $-1, -2, \ldots, -(n+1)$. The claim follows by induction. ■

**Example 27.** Show that any odd degree polynomial with real coefficients has at least one real root.

**Exercise 28** (Putnam 1999 A2)**.** Show that for some fixed positive integer $n$, we can always express a polynomial with real coefficients which is nowhere negative as a sum of the squares of $n$ polynomials.

**Walkthrough** —

**(a)** Show that the real roots of $P$ have even multiplicity.

**(b)** Conclude that $P$ can be expressed as a product of monic quadratic polynomials with real coefficients having nonreal roots, and even powers of linear polynomials with real coefficients.

**(c)** Show that a monic quadratic polynomial with real coefficients having nonreal roots is the sum of the squares of two polynomials with real coefficients.

**Solution 22.** Note that if $P$ is a constant polynomial, then it is clear. Henceforth, let us assume that $P$ is a nonconstant polynomial.

**Claim** — The polynomial $P$ can be written as the product of polynomials, each of which can be expressed as the sum of the squares of two polynomials with real coefficients.

*Proof of the Claim.* Since $P$ has real coefficients, it follows that if $\alpha \in \mathbb{C} \setminus \mathbb{R}$ is a root of $P$, then so is $\overline{\alpha}$. Thus, the nonreal complex roots of $P$ form pairs of complex conjugates. Note that

$$(x-\alpha)(x-\overline{\alpha}) = (x - \mathrm{Re}(\alpha))^2 + \mathrm{Im}(\alpha)^2.$$

Decomposing $P$ over the pairs of nonreal complex conjugate roots, and the real roots, it follows that $P$ can be expressed as the product

$$cf(x) \prod_{a \in A} (x-a)^{m_a},$$

where $c$ denote the leading coefficient of $P$, $f(x)$ denotes the product of (possibly no) quadratic polynomials of the form $(x-a)^2+b^2$ with $a \in \mathbb{R}, b \in \mathbb{R}\setminus\{0\}$, and $A$ denotes the set of real roots of $P$, and for an element $a \in A$, the multiplicity of $a$ is denoted by $m_a$.

Evaluating $P$ at a suitable real number (for instance, at $1+\sum_{a\in A} a$ (resp. 1) if $A$ is nonempty (resp. empty)), it follows that $c > 0$.

Let $a$ be an element of $A$. Since $A$ is finite, there exists a real number $\varepsilon > 0$ such that the interval $(a-\varepsilon, a+\varepsilon)$ contains no real roots of $P$ other than $a$. If $m_a$ were odd, then the sign of $P(x)$ would not remain constant as $x$ ranges over in $(a-\varepsilon, a+\varepsilon)\setminus\{a\}$. Hence, it follows that $m_a$ is even.

Since $c > 0$ amd $m_a$ is even for any $a \in A$, the Claim follows. □

**Claim** — Let $f_1(x), g_1(x), f_2(x), g_2(x)$ be polynomials with real coefficients. Then the following holds.

$$\begin{aligned}&(f_1(x)^2+g_1(x)^2)(f_2(x)^2+g_2(x)^2)\\&=(f_1(x)f_2(x)-g_1(x)g_2(x))^2+(f_1(x)g_2(x)-f_2(x)g_1(x))^2\end{aligned}$$

*Proof of the Claim.* Note that

$$\begin{aligned}&(f_1(x)^2+g_1(x)^2)(f_2(x)^2+g_2(x)^2)\\&=f_1(x)^2f_2(x)^2+g_1(x)^2g_2(x)^2-2f_1(x)f_2(x)g_1(x)g_2(x)\\&\quad+f_1(x)^2g_2(x)^2+f_2(x)^2g_1(x)^2+2f_1(x)g_2(x)f_2(x)g_1(x)\\&=(f_1(x)f_2(x)-g_1(x)g_2(x))^2+(f_1(x)g_2(x)+f_2(x)g_1(x))^2.\end{aligned}$$

□

Combining the above Claims, and using induction, the result follows. ■

**Exercise 29** (RMO 2017a P3, ). Let $P(x)=x^2+\frac{x}{2}+b$ and $Q(x)=x^2+cx+d$ be two polynomials with real coefficients such that $P(x)Q(x)=Q(P(x))$ for all real $x$. Find all real roots of $P(Q(x))=0$.

**Solution 23.** Let $\alpha$ be a root of $P(x)$ in $\mathbb{C}$. Substituting $x=\alpha$ in $P(x)Q(x)=Q(P(x))$, we obtain $Q(0)=0$, which shows that $d=0$. This gives

$$P(x)(x^2+cx)=P(x)(P(x)+c),$$

which yields $P(x)+c=x^2+cx$. It follows that $c=\frac{1}{2}, b=-\frac{1}{2}$.

Note that the roots of $P(x)$ are $-1, \frac{1}{2}$. Hence, any root $\beta$ of $P(Q(x))=0$ satisfies $Q(\beta)=-1$ or $Q(\beta)=\frac{1}{2}$. Since the discrimint of $Q(x)+1$ is negative, it follows that if $\beta$ is a real root of $P(Q(x))=0$, then $Q(\beta)=\frac{1}{2}$, which gives $\beta=-1$ or $\beta=1/2$.

Note that $P(\frac{1}{2})=0$ and $-1, \frac{1}{2}$ are the roots of $Q(x)=1/2$. This shows that the real roots of $P(Q(x))=0$ are $-1, \frac{1}{2}$. ■

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